# **United** International **University**

***School of Science and Engineering***

Mid Term Examination Trimester: Fall-2023

Course Title: Fundamental Calculus

Course Code: Math 1151 Marks: 30 Time: 1 Hour 45 Mins

**Answer <u>all</u> the questions. Answer all parts of a question together.**

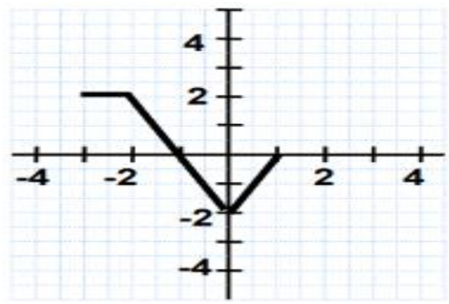
**1. (a)** **Draw** the graph of the following functions and **find** their domain and range. **[5]**

**(i)** $y = 2 - 2x - x^2$ **(ii)** $y = 1 - \sin 2x$

**(b)** The graph of $f(x)$ is given. Use it to **sketch** the graph of the following functions. **[5]**

**(i)** $2 + f(1 - x)$

**(ii)** $1 - 2\left|f(\frac{x}{2})\right|$




**2. (a)** **Evaluate** $f(-5)$, $f(-1)$, and $f(3)$ for the piecewise defined function. Then **sketch** the graph of the function. **[3]**

$$f(x) = \begin{cases} 2; & x < -3 \\ \sqrt{9 - x^2}; & -3 \le x < 3 \\ x - 2; & x \ge 3 \end{cases}$$

**(b)** Use the table to evaluate the following expressions: **[3]**

| $x$ | -3 | -1 | 2 | 5 |
|---|---|---|---|---|
| $f(x)$ | 7 | 2 | -1 | -1 |
| $g(x)$ | 9 | -3 | -8 | 2 |

**(i)** $(f \circ g)(-1)$ **(ii)** $(g \circ g)(5)$ **(iii)** $(g \circ f)(2)$

**(c)** The graph of $f(x)$ is given. **[4]**

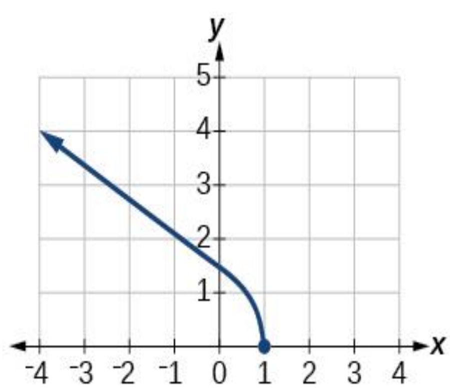
**(i)** **Determine** whether $f(x)$ is one to one function, or not.

**(ii)** **Complete** the following table.

| $x$ | 0 | 2 | 4 |
|---|---|---|---|
| $f^{-1}(x)$ | | | |

**(iii)** **Sketch** the graph of $f^{-1}(x)$ along with $f(x)$.

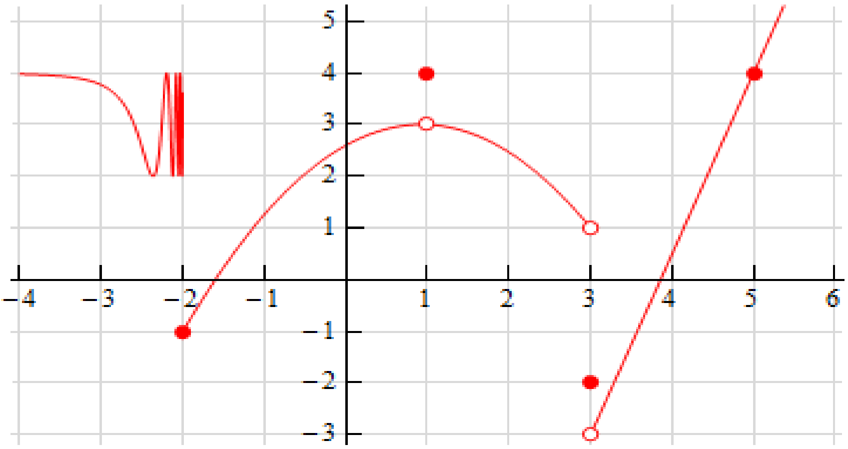
**(iv)** **What** is the domain and range of $f^{-1}(x)$?




**Please Turn Over**

3. **(a)** **Find** the inverse of $f(x) = 2 + 3^{-x}$, **draw** the graph of $f(x)$ and **its inverse** in the same diagram. Also, **state** the domain and range of the inverse function. **[3]**

**(b)** The graph of the function $y = f(x)$ is given. **[5]**




From the figure **write** the answers of the following questions:

**(i)** $\lim_{x\to-2^-} f(x)$ and $\lim_{x\to3^+} f(x)$.

**(ii)** $\lim_{x\to1} f(x)$.

**(iii)** $f(1)$ and **state** the horizontal asymptote(s) of $f(x)$.

**(iv)** **Check** and **explain** the continuity of $f(x)$ at $x = -2$ and $5$.

**(c)** **Find** value of the constant $k$, if possible, that will make the function $f(x)$ continuous everywhere. **[2]**

$$f(x) = \begin{cases} x^3 + 2k; & x > -1 \\ x + 5; & x \leq -1 \end{cases}$$

***BEST OF LUCK!!!***